

Technical Report – Turtle Games

Introduction

Turtle Games is a game manufacturer and retailer with a global customer base. Through the company's sales and customer review data, Turtle Games has a business objective of improving overall sales performance by analysing and considering customer trends. By analysing the company's customer data, predictions on the company's loyalty points scheme can be made. Multiple machine learning techniques have been levied through Python and R to ascertain a deeper picture of the customer data, and therefore predictions can be made to help the marketing team achieve their goals of targeting specific customers, and improve overall sales performance. The machine learning techniques used in this analysis are as follows:

- Linear and multiple linear regression
- Logistic regression with decision trees
- K-means clustering
- Natural language processing for sentiment analysis

Data wrangling

To begin the analysis; specific libraries were implemented in Python and R. The libraries in Python were Pandas and Numpy for the creation of DataFrames and working with data, Matplotlib, Seaborn and other visualisation libraries, and sklearn and statsmodels for machine learning and statistical analysis. For R, the tidyverse library was opened to gain access to tidyr, dplyr and ggplot2, alongside DataExplorer for analysis reports, and moments for statistical analyses.

The csv file was read in each programming language, and the data was wrangled to make it more useful for the analysis.

The wrangling process involved removing redundant columns, and changing column names to be more representative of the metadata. In R, the product column was changed from a continuous integer column to a factor. This was because R implicitly inferred the product column to be continuous data, rather than categorical, as they were numeric product codes.

Exploratory descriptive analysis visualisations were created using ggplot2 with R, which will be used in the presentation for senior stakeholders at Turtle Games (figure 1 and 2).

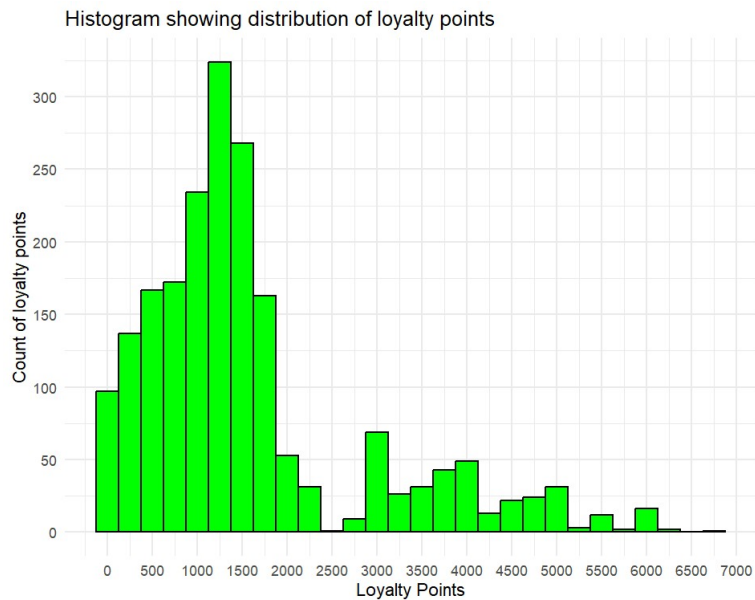


Figure 1 - Histogram showing distribution of loyalty points

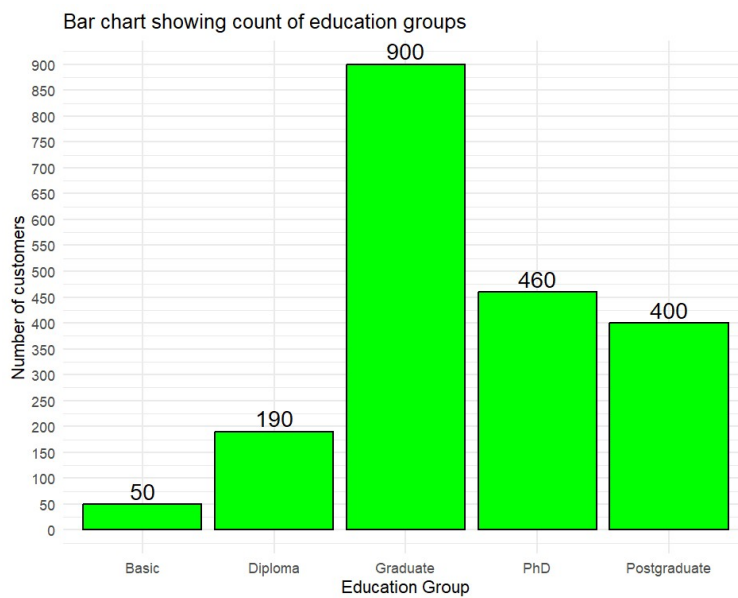


Figure 2 - Bar chart showing count of education group

Before the statistical analysis, separate DataFrames in Python were created for each specific machine learning method.

Linear regression and multiple linear regression.

Linear regression was undertaken through Python comparing loyalty points as the dependent variable, and age, spending score, and income as the individual independent variables. Using Ordinary Least Squares regression, results were achieved for each linear regression (figures 3, 4 and 5). Spending score had the most impact on

loyalty points by 45.2% - the R^2 value was 0.452. Visualisations were created for each linear regression model (figures 6, 7 and 8).

OLS Regression Results						
Dep. Variable:	y		R-squared:	0.452		
Model:	OLS		Adj. R-squared:	0.452		
Method:	Least Squares		F-statistic:	1648.		
Date:	Sun, 19 Jul 2026		Prob (F-statistic):	2.92e-263		
Time:	15:49:11		Log-Likelihood:	-16550.		
No. Observations:	2000		AIC:	3.310e+04		
Df Residuals:	1998		BIC:	3.312e+04		
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-75.0527	45.931	-1.634	0.102	-165.129	15.024
x	33.0617	0.814	40.595	0.000	31.464	34.659
Omnibus:	126.554	Durbin-Watson:	1.191			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	260.528			
Skew:	0.422	Prob(JB):	2.67e-57			
Kurtosis:	4.554	Cond. No.	122.			

Figure 3 - OLS Regression Results for spending vs loyalty points

OLS Regression Results						
Dep. Variable:	y		R-squared:	0.380		
Model:	OLS		Adj. R-squared:	0.379		
Method:	Least Squares		F-statistic:	1222.		
Date:	Sun, 19 Jul 2026		Prob (F-statistic):	2.43e-209		
Time:	15:49:11		Log-Likelihood:	-16674.		
No. Observations:	2000		AIC:	3.335e+04		
Df Residuals:	1998		BIC:	3.336e+04		
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-65.6865	52.171	-1.259	0.208	-168.001	36.628
x	34.1878	0.978	34.960	0.000	32.270	36.106
Omnibus:	21.285	Durbin-Watson:	3.622			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	31.715			
Skew:	0.089	Prob(JB):	1.30e-07			
Kurtosis:	3.590	Cond. No.	123.			

Figure 4 - OLS Regression Results for income vs loyalty points

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.002			
Model:	OLS	Adj. R-squared:	0.001			
Method:	Least Squares	F-statistic:	3.606			
Date:	Sun, 19 Jul 2026	Prob (F-statistic):	0.0577			
Time:	15:49:11	Log-Likelihood:	-17150.			
No. Observations:	2000	AIC:	3.430e+04			
Df Residuals:	1998	BIC:	3.431e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	1736.5177	88.249	19.678	0.000	1563.449	1909.587
x	-4.0128	2.113	-1.899	0.058	-8.157	0.131
Omnibus:	481.477	Durbin-Watson:	2.277			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	937.734			
Skew:	1.449	Prob(JB):	2.36e-204			
Kurtosis:	4.688	Cond. No.	129.			

Figure 5 - OLS Regression Results for age vs loyalty points

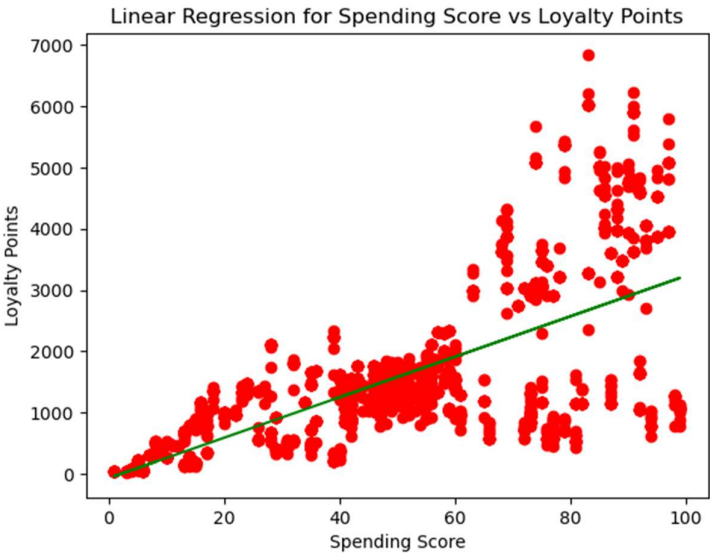


Figure 6 - Linear regression of spending score vs loyalty points

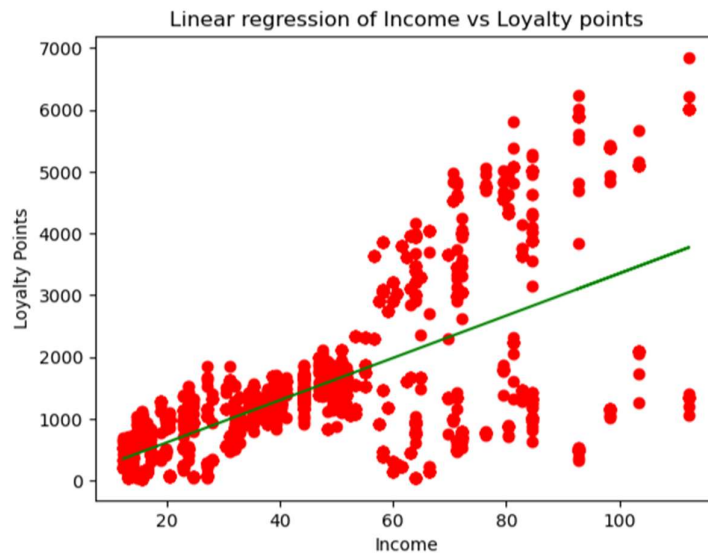


Figure 7 - Linear regression for income vs loyalty points

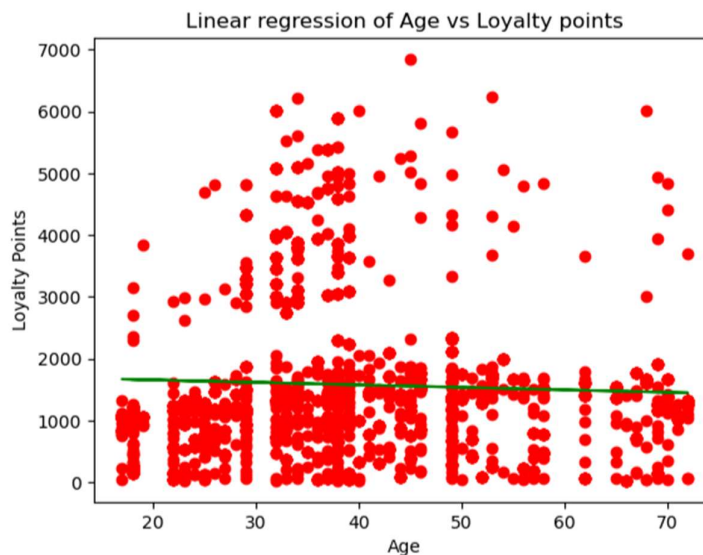


Figure 8 - Linear regression for age vs loyalty points

Multiple linear regression was completed through R. Each numeric variable was evaluated for normality. First by creating QQplots (see figures 9, 10, 11 and 12), and then by using the Shapiro-Wilk test. Every variable showed a p -value that was significantly less than 0.05. However, the skewness and kurtosis of the variables were evaluated and showed that even though they were not exhibiting statistical normality, they were exhibiting practical normality (figures 13, 14, 15 and 16).

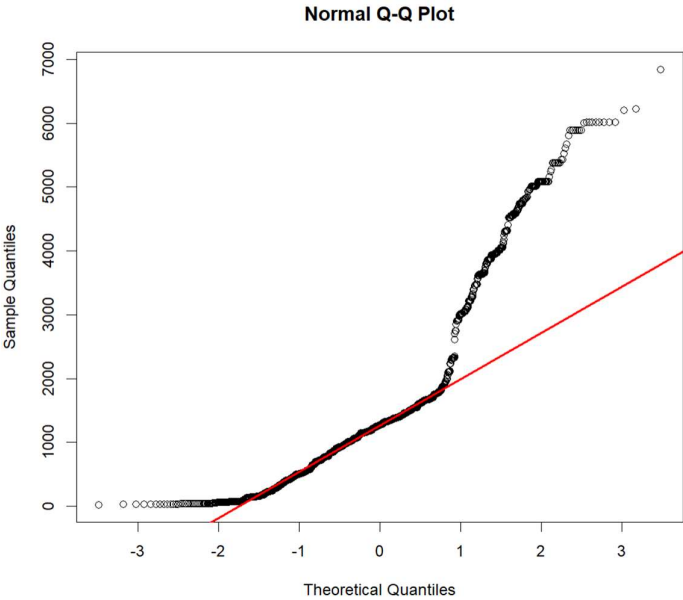


Figure 9 - QQplot for loyalty points

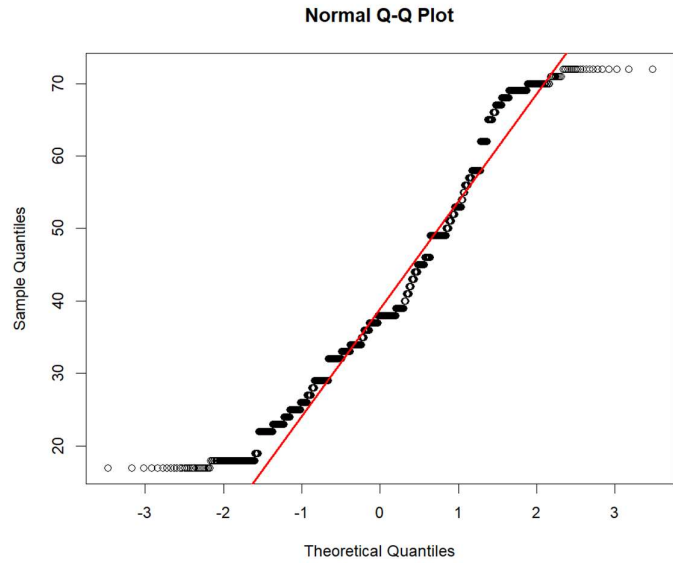


Figure 10 - QQplot for age

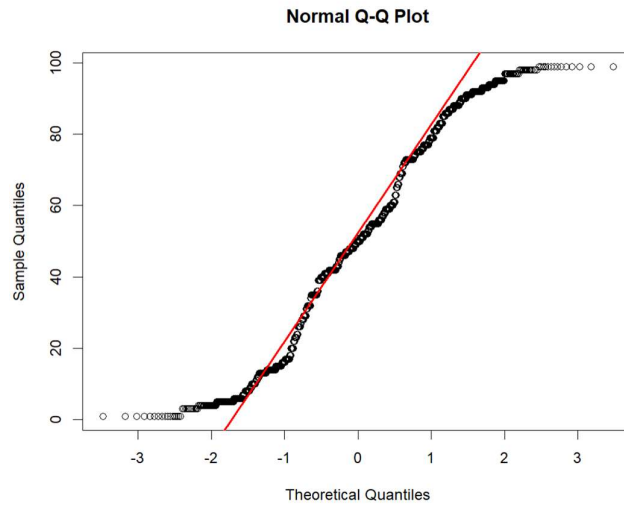


Figure 11 - QQplot for spending score

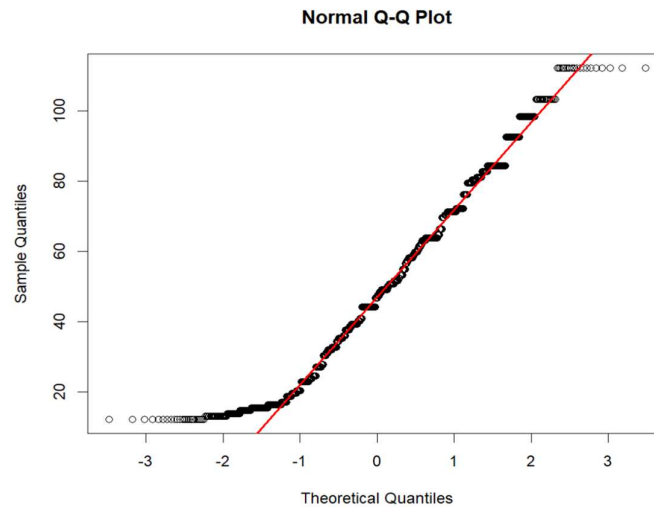


Figure 12 - QQplot for income

```
> shapiro.test(turtle1$loyalty_points)
```

Shapiro-wilk normality test

```
data:  turtle1$loyalty_points  
W = 0.84307, p-value < 2.2e-16
```

```
> skewness(turtle1$loyalty_points)
```

```
[1] 1.463694
```

```
> kurtosis(turtle1$loyalty_points)
```

```
[1] 4.70883
```

Figure 13 - RStudio console output for Shapiro-Wilk test, skewness and kurtosis for loyalty points

```
> shapiro.test(turtle1$age)

      Shapiro-Wilk normality test

data:  turtle1$age
W = 0.95242, p-value < 2.2e-16

> skewness(turtle1$age)
[1] 0.6088437
> kurtosis(turtle1$age)
[1] 2.80893
```

Figure 14 - RStudio console output for Shapiro-Wilk test, skewness and kurtosis for age

```
> shapiro.test(turtle1$spending_score)

      Shapiro-Wilk normality test

data:  turtle1$spending_score
W = 0.96835, p-value < 2.2e-16

> skewness(turtle1$spending_score)
[1] -0.04161713
> kurtosis(turtle1$spending_score)
[1] 2.110333
```

Figure 15 - RStudio console output for Shapiro-Wilk test, skewness and kurtosis for spending score

```
> shapiro.test(turtle1$income)

      Shapiro-Wilk normality test

data:  turtle1$income
W = 0.96768, p-value < 2.2e-16

> skewness(turtle1$income)
[1] 0.412842
> kurtosis(turtle1$income)
[1] 2.591949
```

Figure 16 - RStudio console output for Shapiro-Wilk test, skewness and kurtosis for income

Multiple linear regressions were completed with loyalty points as the dependent variable and varying combinations of age, income, and spending score as the dependent variables. Completing a multiple linear regression for all of the dependent variables resulted in an adjusted R^2 value of 0.8397. This states that age, income, and spending score indicate 84% of the loyalty points (figure 17).

```
call:
lm(formula = loyalty_points ~ age + income + spending_score,
    data = turtle_num)

Residuals:
    Min       1Q   Median       3Q      Max
-1819.11  -350.84    4.61   291.00  1894.62

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2203.0598    52.3609  -42.08  <2e-16 ***
age             11.0607     0.8688   12.73  <2e-16 ***
income         34.0084     0.4970   68.43  <2e-16 ***
spending_score  34.1832     0.4519   75.64  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 513.8 on 1996 degrees of freedom
Multiple R-squared:  0.8399,    Adjusted R-squared:  0.8397
F-statistic: 3491 on 3 and 1996 DF,  p-value: < 2.2e-16
```

Figure 17 - RStudio console output for multiple linear regression summary

The QQplot was made for the residuals, alongside the residual vs fitted values plot. The QQplot of the residuals showed that the values followed the reference line and indicated normality (figure 18). However, the residuals vs fitted values plot did not show a random scattering around zero, but rather a U shape (figure 19). Perhaps this indicates a squared rather than linear relationship. Transforming some (or all) of the variables through square rooting would create a better fitted model. Nevertheless, predictions of the model were completed due to the confidence given from the relatively high R^2 value.

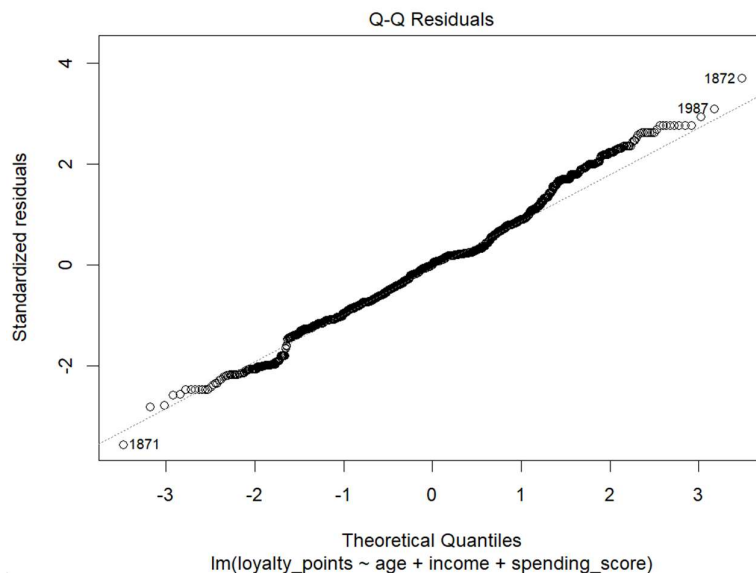


Figure 18 - QQplot for residuals of multiple linear regression model

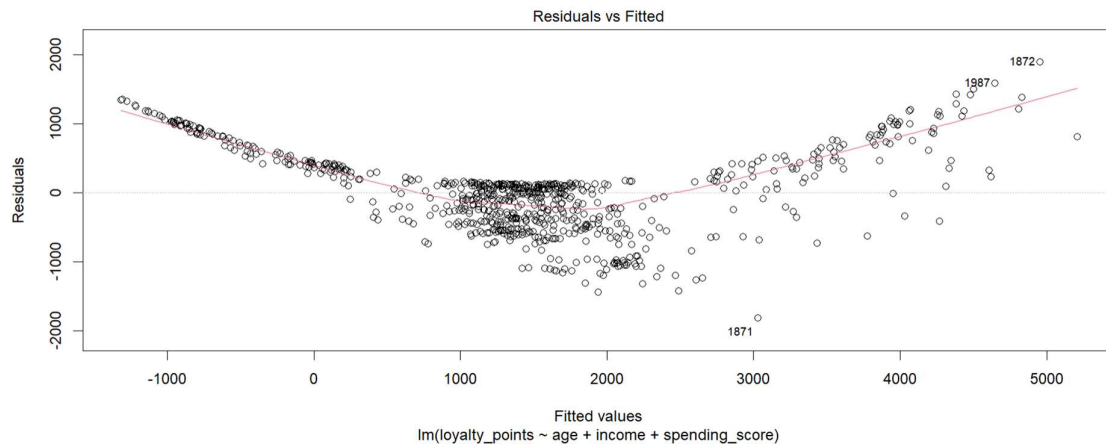


Figure 19 - Residuals vs Fitted plot of multiple linear regression model

Four predictions were made (figure 20). The synthetic data was created with the knowledge gained from decision tree regression. The predictions proved the decision tree regression, which is that spending score and then income have the most input towards loyalty points. Nevertheless, each of the X variables for the multiple linear regression had an influence on the prediction.

	age	spending_score	income	predicted_loyalty_points
1	45	50	40	1364.165
2	50	65	49	2238.292
3	55	70	43	2260.461
4	60	85	70	3746.738

Figure 20 – data frame showing predictions on synthetic data using the multiple linear regression model

Decision tree regressor

To create the decision tree, the data was split into a 70:30 for training and testing. The decision tree was created using the training data, which showed it to be exceptionally large (figure 21). The accuracy of the tree was evaluated and a decision to prune the tree to two branches was made (figure 22). A second tree with two branches was created (figure 23). The pruned tree was evaluated using the test data and predicted value, which resulted in Mean Absolute Error, Mean Squared Error and Root Mean Squared Error being similar to the larger tree (figure 24). This showed that the tree being pruned to two branches was a success. A final tree was created (figure 25) on the whole data set to ascertain specific factors for accumulating loyalty points. The final model showed that customers with a spending score greater than 67 was the greatest factor for higher loyalty points, followed by an income greater than £43870.

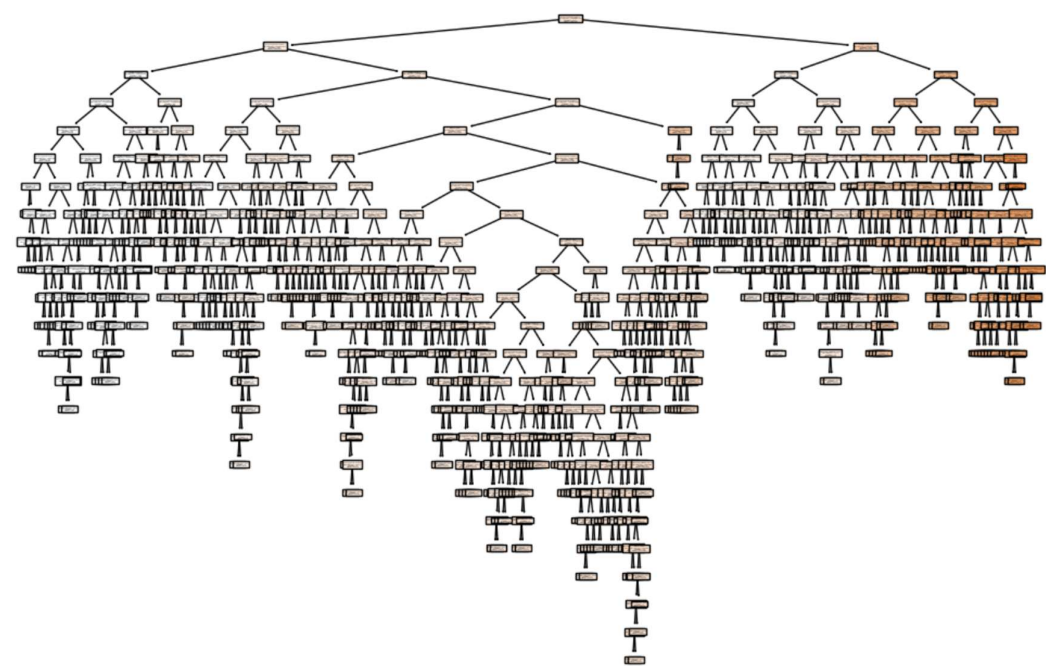


Figure 21- Unpruned decision tree on training data

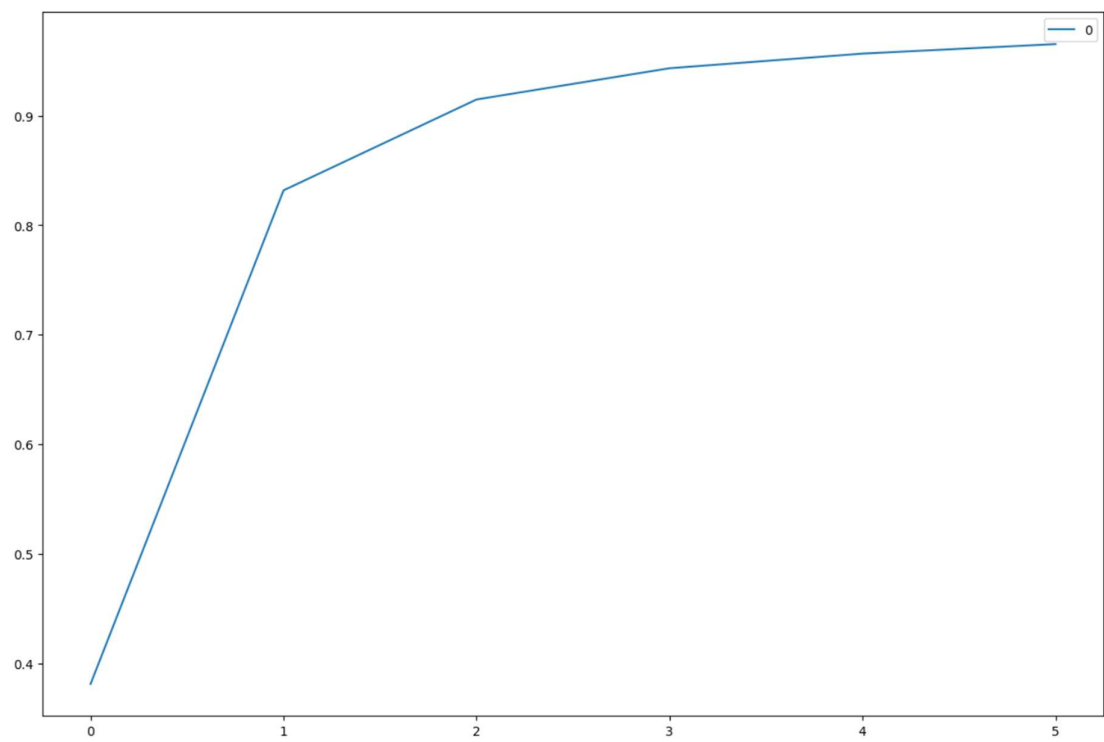


Figure 22 - plot of accuracy values for decision tree

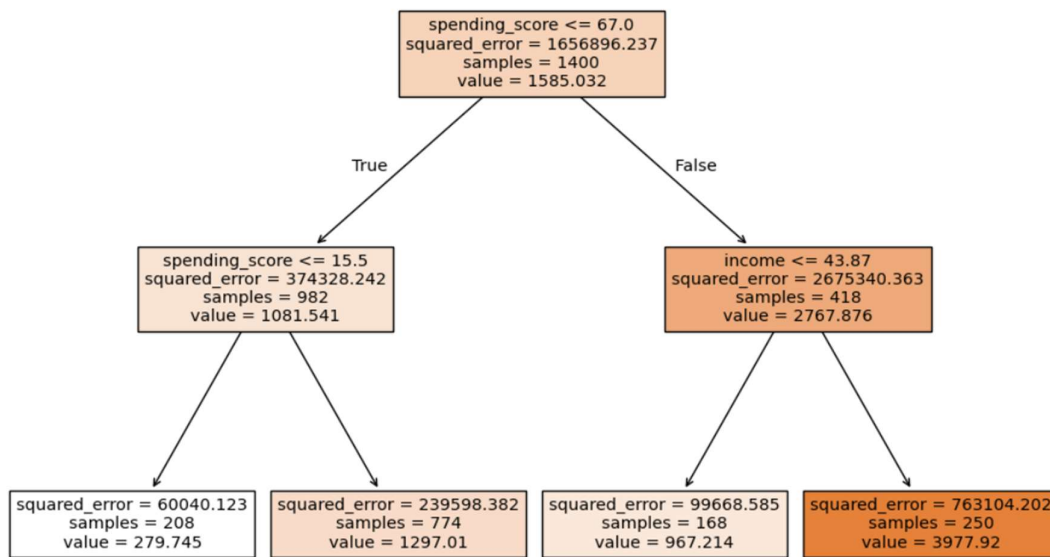


Figure 23 - Pruned decision tree on training data

```

# Evaluate the pruned tree.
# Evaluate the model.
# Predict the response for the data test.
y_predict_pruned = regressor.predict(X_test)

# Specify to print the MAE and MSE (to evaluate the accuracy of the new model).
print("Mean Absolute Error: ", metrics.mean_absolute_error(y_test, y_predict_pruned))
print("Mean Squared Error: ", metrics.mean_squared_error(y_test, y_predict_pruned))
# Calculate the RMSE.
print("Root Mean Squared Error: ", math.sqrt(metrics.mean_squared_error(y_test, y_predict_pruned)))

Mean Absolute Error: 32.51333333333333
Mean Squared Error: 8072.756666666667
Root Mean Squared Error: 89.84852067044102

# Round the pruned tree errors.
print("Mean Absolute Error: ", round(metrics.mean_absolute_error(y_test, y_predict_pruned), 2))
print("Mean Squared Error: ", round(metrics.mean_squared_error(y_test, y_predict_pruned), 2))
# Calculate the RMSE.
print("Root Mean Squared Error: ", round(math.sqrt(metrics.mean_squared_error(y_test, y_predict_pruned)), 2))

Mean Absolute Error: 32.51
Mean Squared Error: 8072.76
Root Mean Squared Error: 89.85
  
```

The errors are the same with the original tree and the pruned tree. The model should be used at depth 2 for a simpler model.

Figure 24 - Python code evaluating errors of pruned tree.

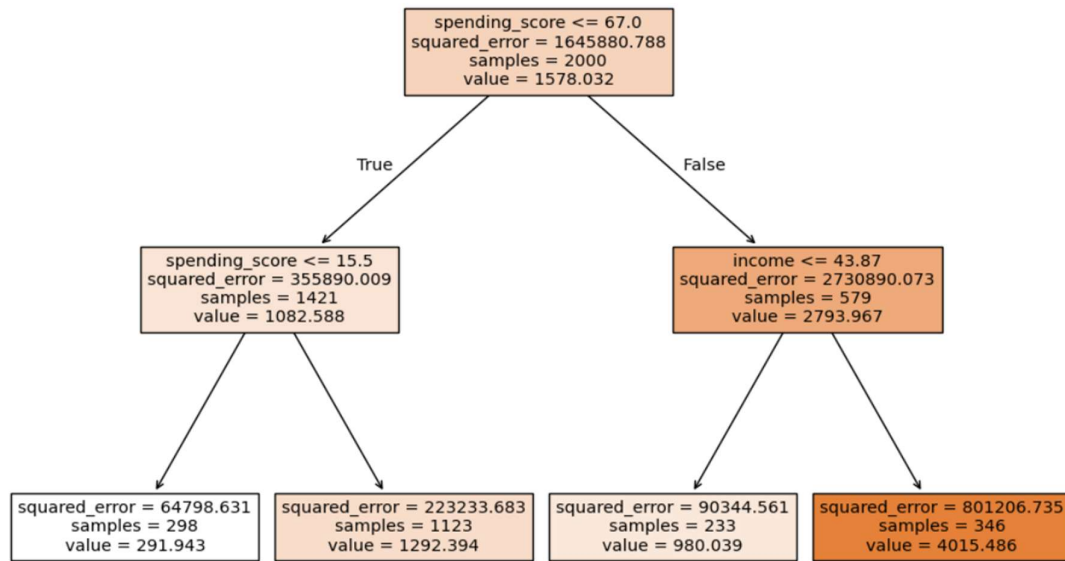


Figure 25 - Final decision tree on all of the data

K-Means clustering

K-Means clustering was completed comparing income with spending score. Visualisations were created to gain an initial insight into the data. The scatterplot showed some distinct groups, and the density plots showed multiple peaks, which suggested multiple groups also (figure 26).

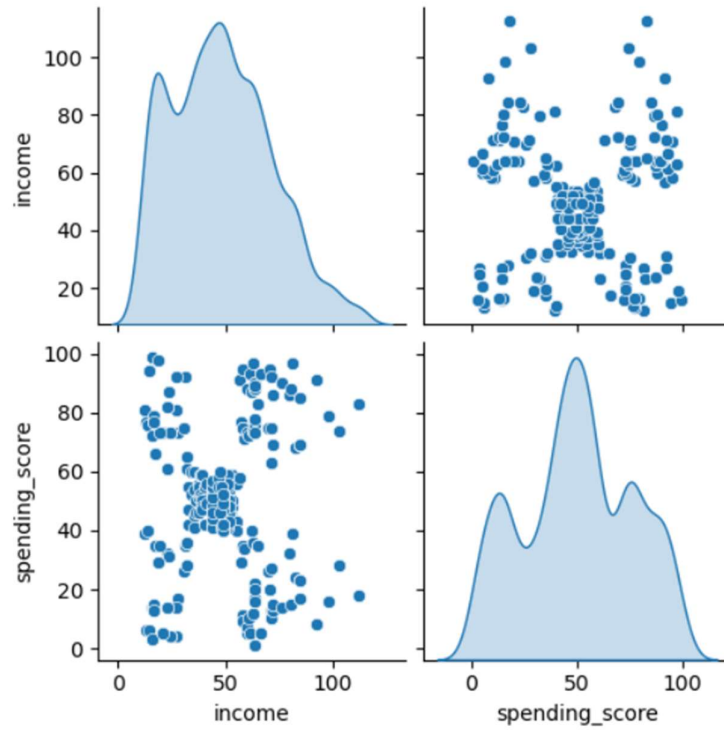


Figure 26 - Pairplot of scatterplots and density plots comparing income and spending score

The silhouette and elbow methods were used on the two variables to ascertain how many distinct groups there would be in the data (figures 27 and 28). The methods showed options between 3 and 5, so clusters were created for each option and evaluated using scatterplots and density plots again. Clustering the data into five groups proved to be most successful (figure 29).

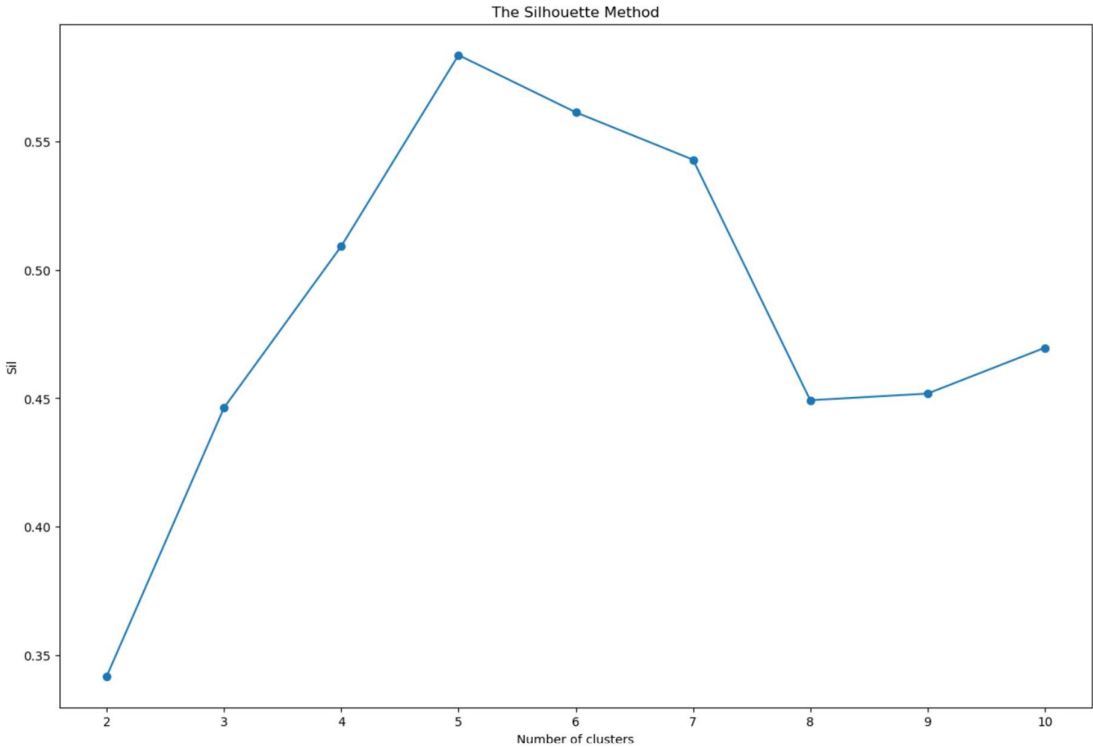


Figure 27 - Silhouette method plot for *k*-means clustering

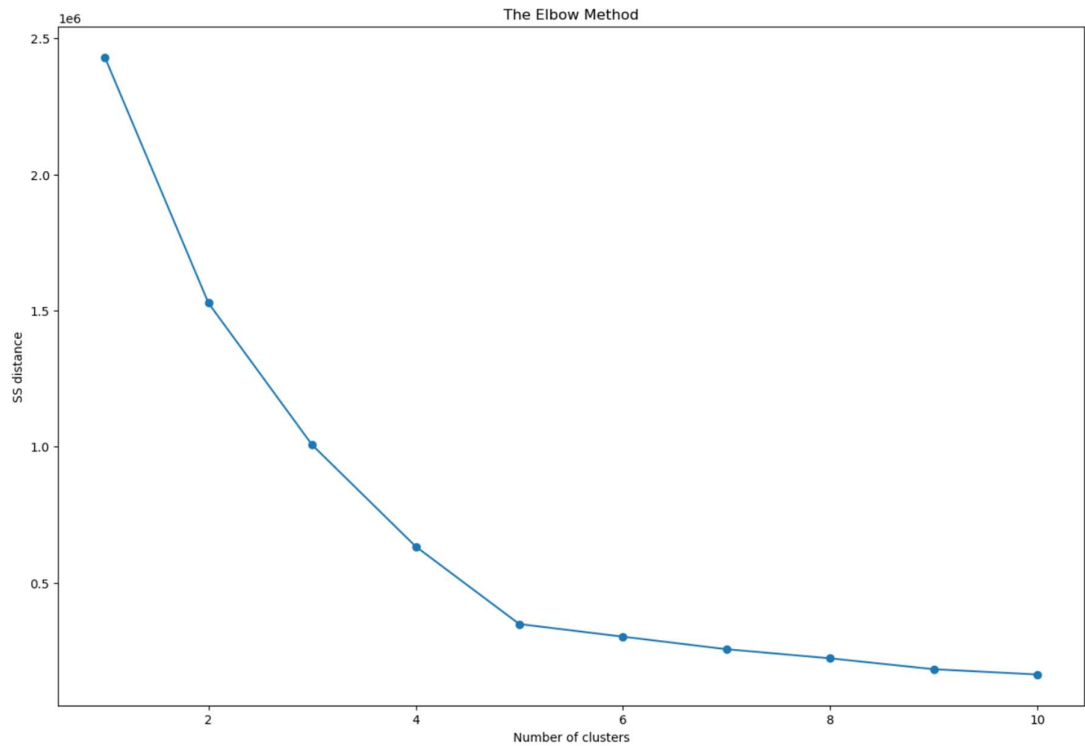


Figure 28 - Elbow method plot for *k*-means clustering

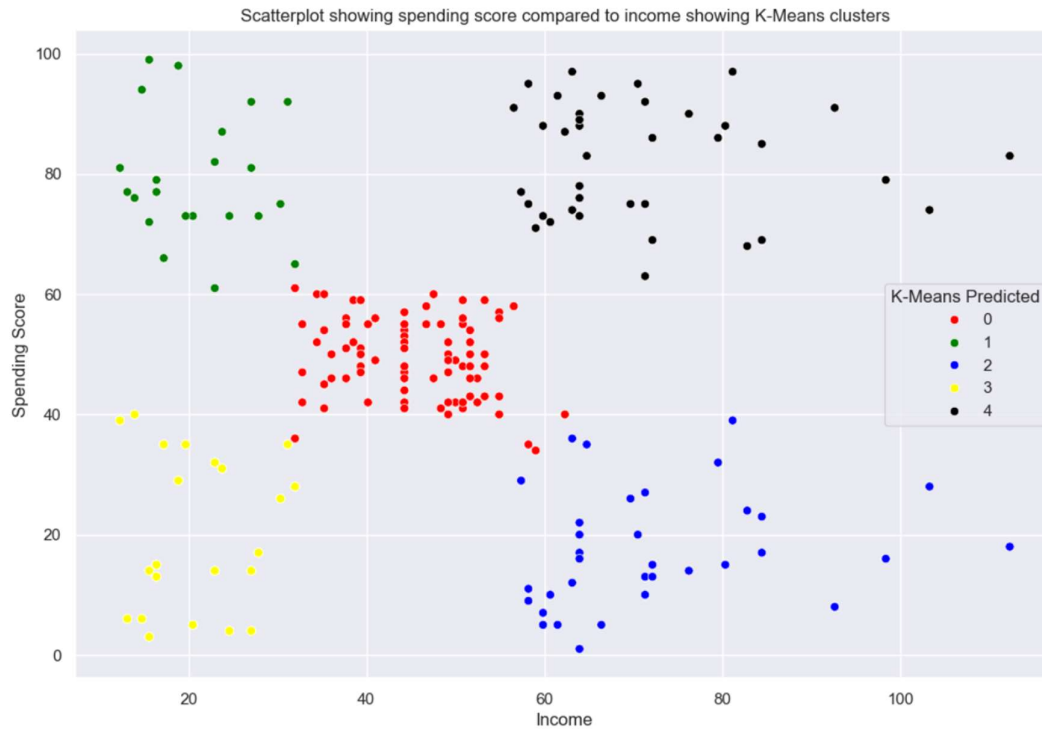


Figure 29 - Scatterplot showing five clusters

Descriptive analyses for each individual group was completed.

Sentiment analysis

Sentiment analysis was completed on the reviews and summary columns of the data using natural language processing. Each word of every review and summary was created into their own individual token. The stop words were taken out to leave the more appropriate words. A frequency distribution of the word tokens was completed to ascertain the 15 most common words for reviews and columns (figures 30 and 31). This showed that there are customers using the summary section as a place to put star ratings.

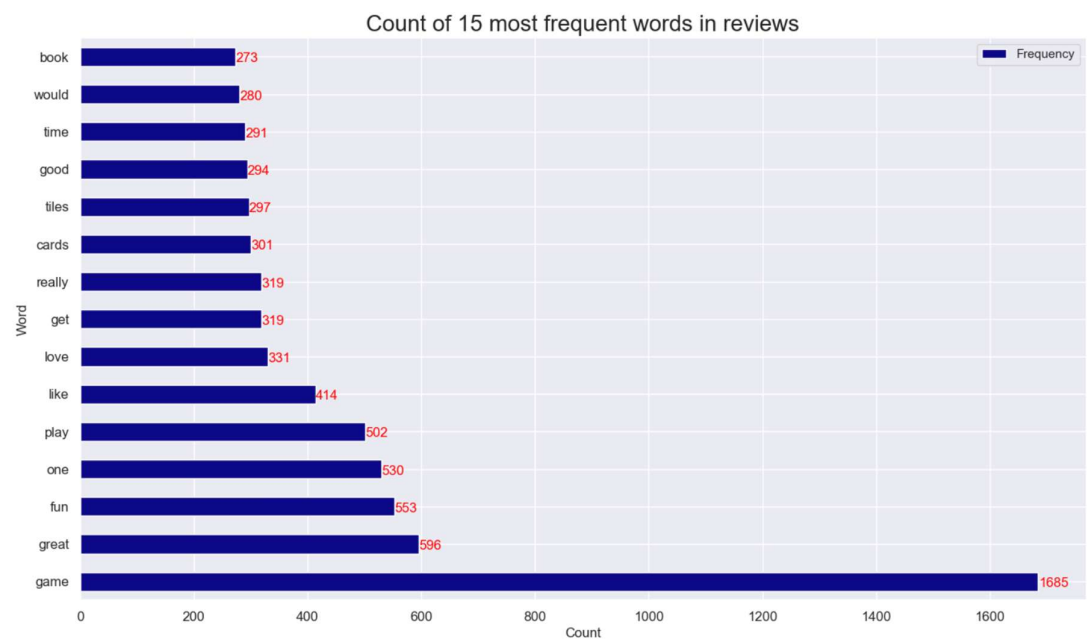


Figure 30 - Bar plot showing 15 most frequent words in reviews

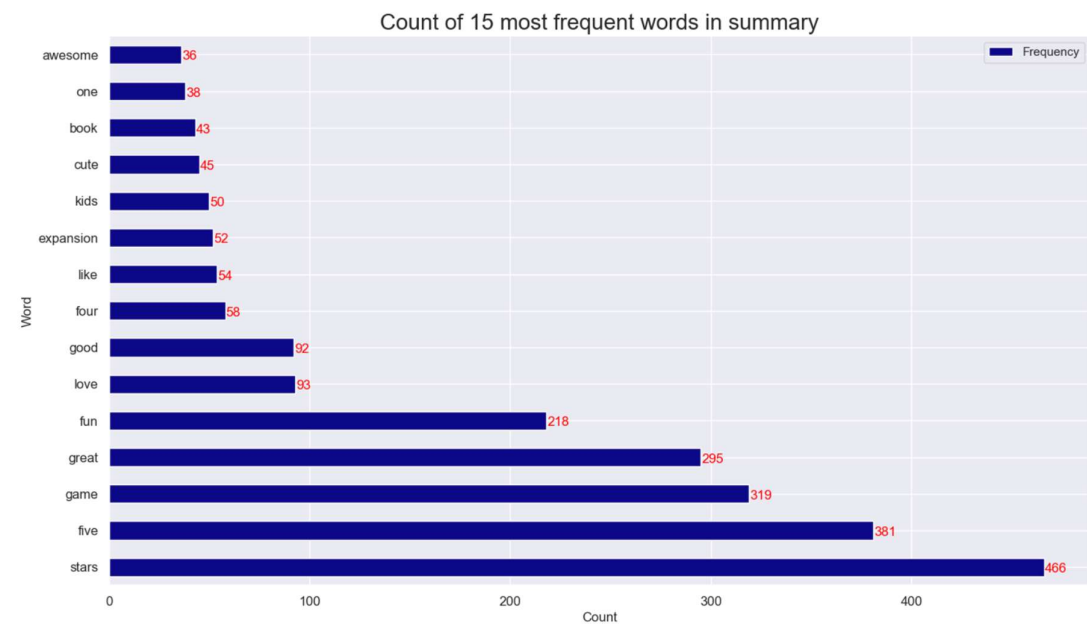


Figure 31 - Bar plot showing 15 most frequent words in summary

Sentiment analysis was completed on every review and summary individually to determine the polarity and subjectivity, using the TextBlob package. The results of the sentiment analysis for review and summary were put into histograms. This showed a slightly positive polarity score for review and a relative neutral polarity score for summary. For subjectivity, the histograms showed a balanced subjectivity score for reviews, and a very objective score for summary (figures 32, 33, 34 and 35). This shows that the two columns are being used correctly.

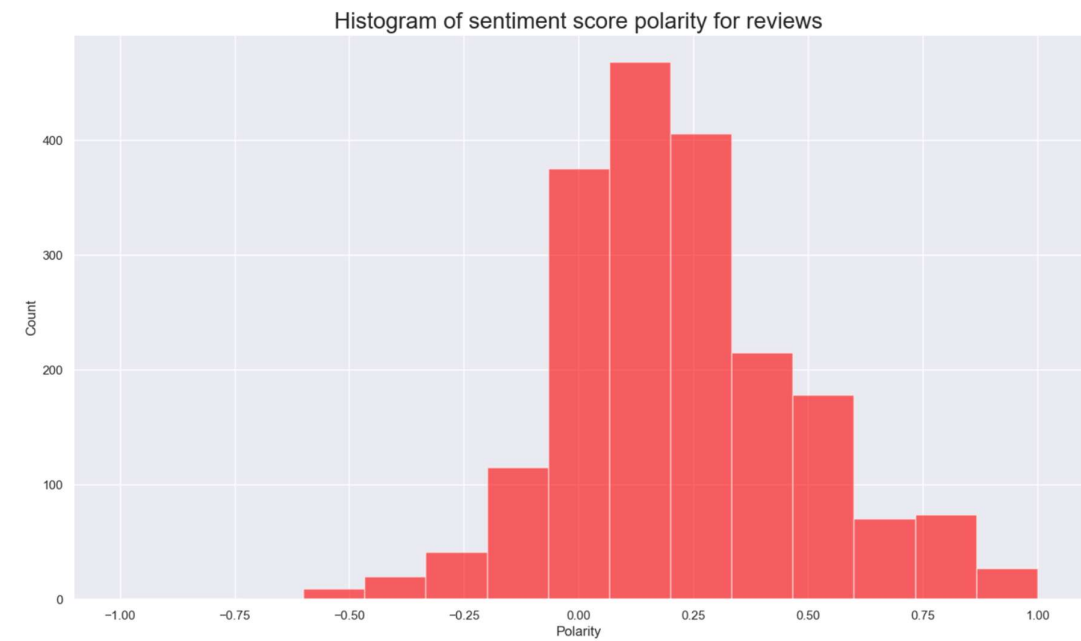


Figure 32 - Histogram showing the distribution of polarity scores for reviews

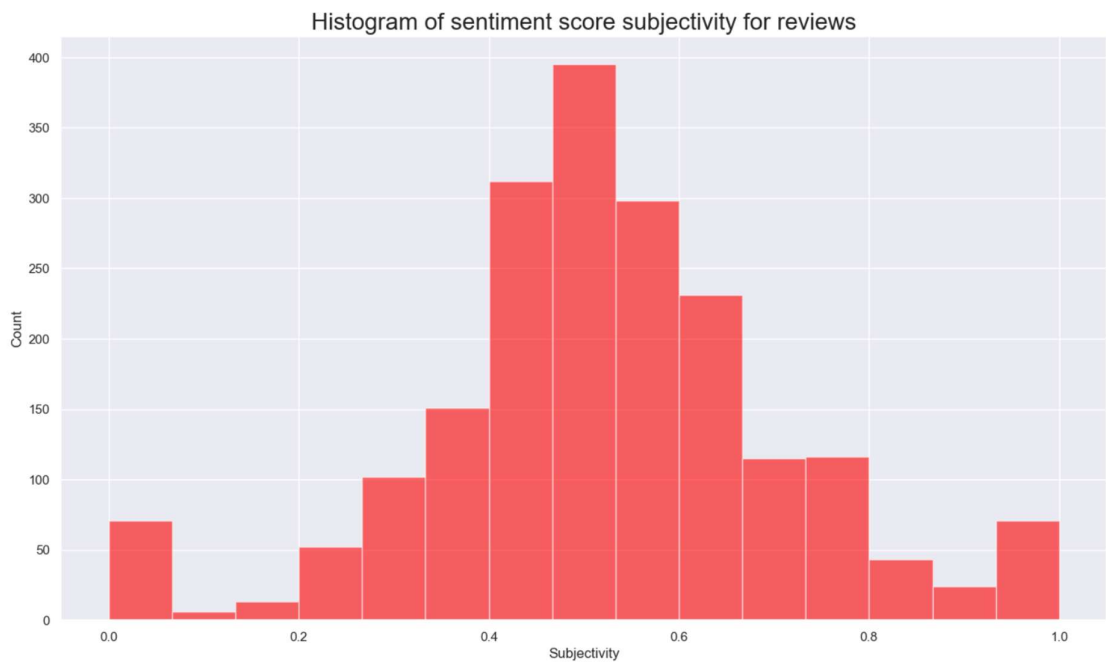


Figure 33 - Histogram showing distribution of subjectivity scores for reviews

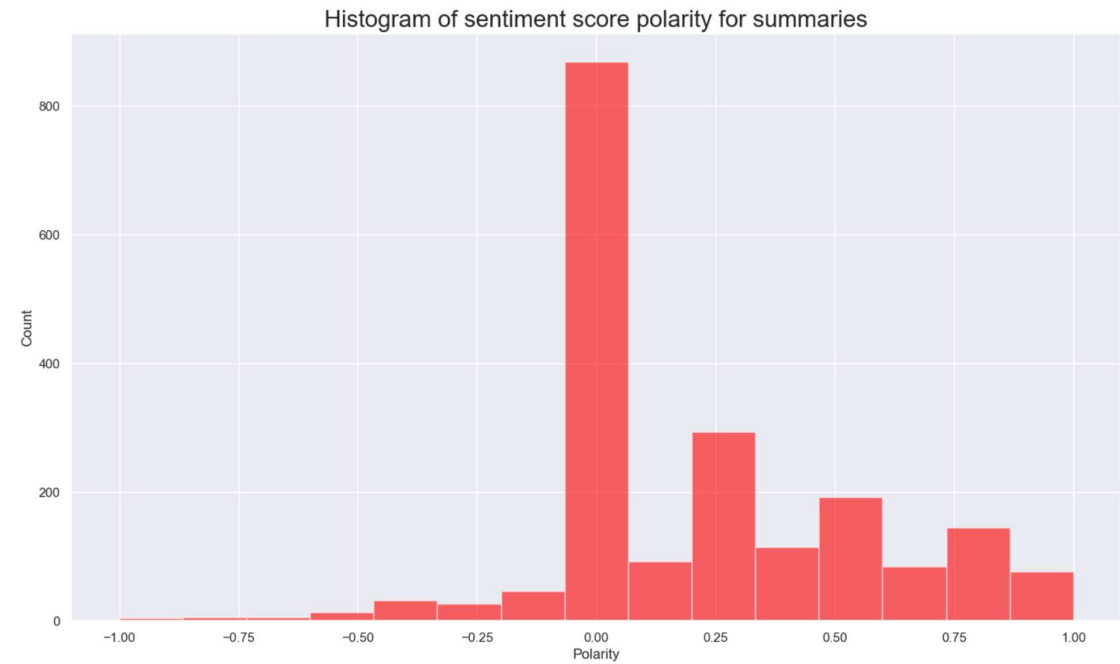


Figure 34 - Histogram showing polarity scores for summary

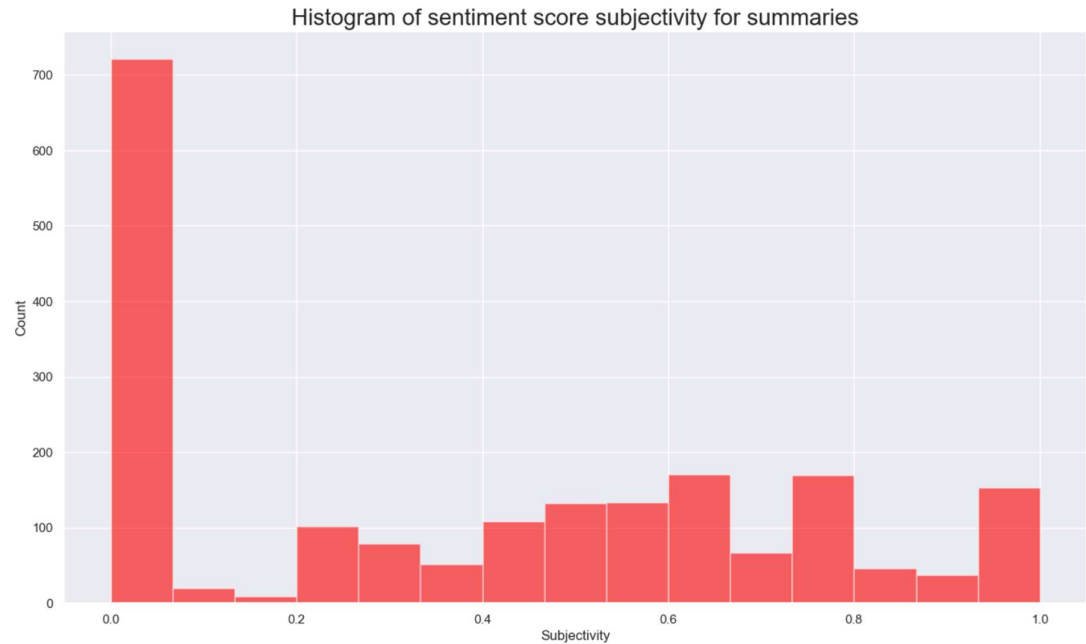


Figure 35 - Histogram showing subjectivity score for summary

Insights

The insights from this analysis are such that the greater age, income, and spending score will determine greater loyalty points. More specifically, a customer with a spending score greater than 67, and an income greater than £43870 will accumulate the most loyalty points. There are five distinct groups of customers in the data when looking

at income and spending score. Finally, the sentiment analysis shows that the customers are on average positive of the products they receive from Turtle Games.

Recommendations.

When looking at the clusters created, there is a group of customers with relative high income, but low spending score. The marketing team should aim for this high earning group, perhaps with deals and offers, to encourage them to improve their spending score and therefore improve their loyalty towards Turtle Games. This will result in growth in the company, and ultimately more revenue and profit.

Using the positive remarks of genuine customer reviews is a great marketing strategy to improve sales performance. If the reviews are targeted at the aforementioned group of customers, this may encourage them to engage with Turtle Games more frequently and improve the loyalty score.

Ultimately having more customers loyal to the brand is a great way of improving overall sales performance.

Appendix

Figure 1 – Histogram showing distribution of loyalty points

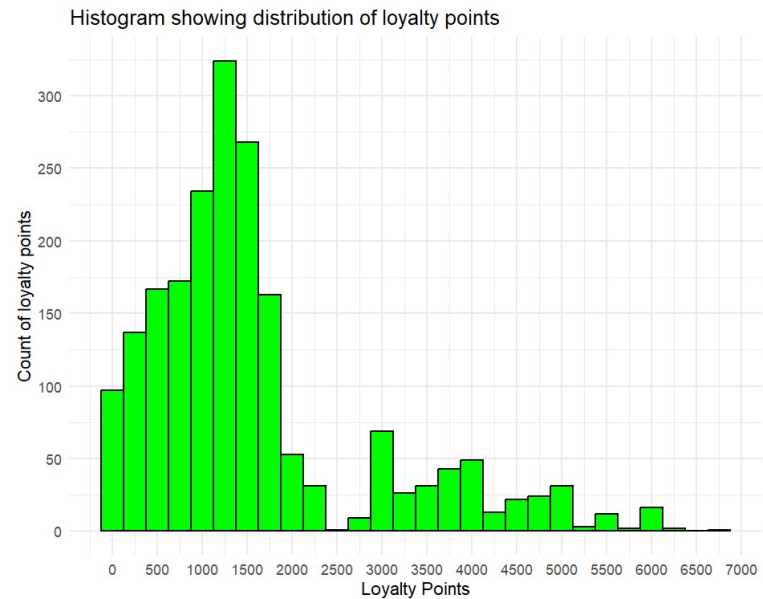


Figure 2 – Bar chart showing count of education groups

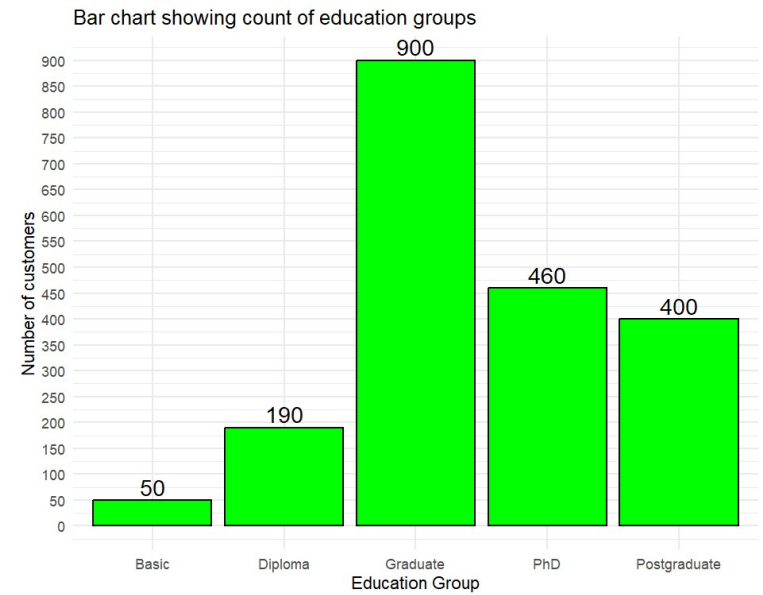


Figure 3 – OLS Regression Results for spending vs loyalty

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Omnibus:	126.554	Durbin-Watson:	1.191			
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Figure 5 – OLS Regression Results for spending vs loyalty points

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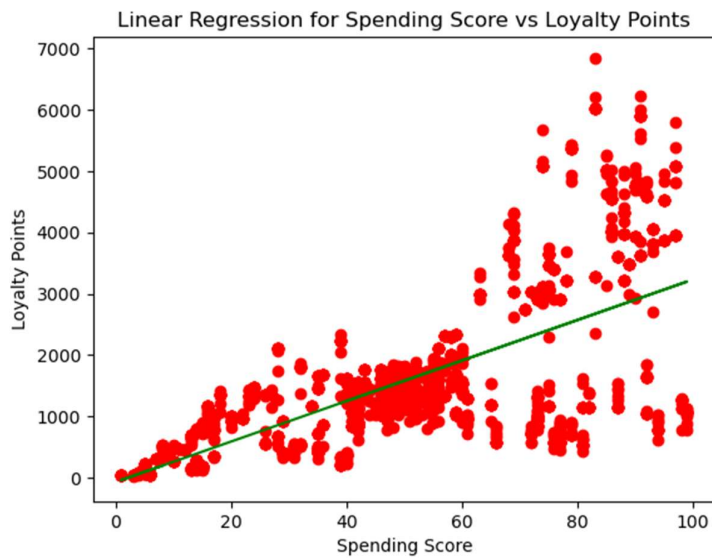


Figure 7 – Linear regression for income vs loyalty points

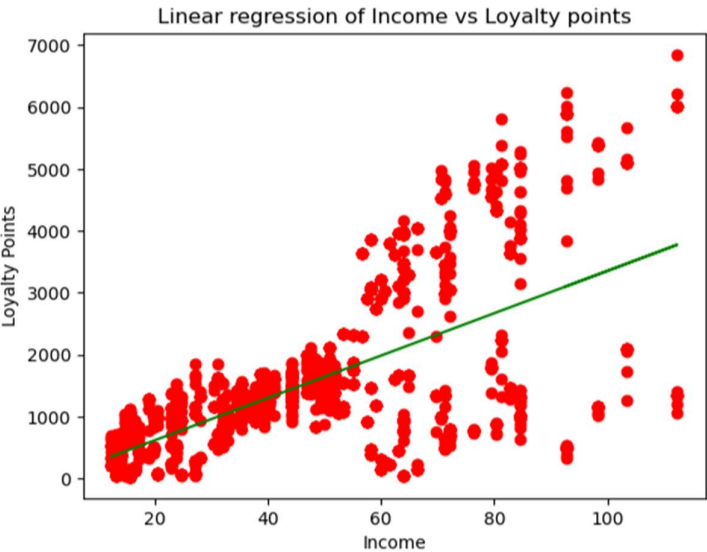


Figure 8 – Linear regression for age vs loyalty points

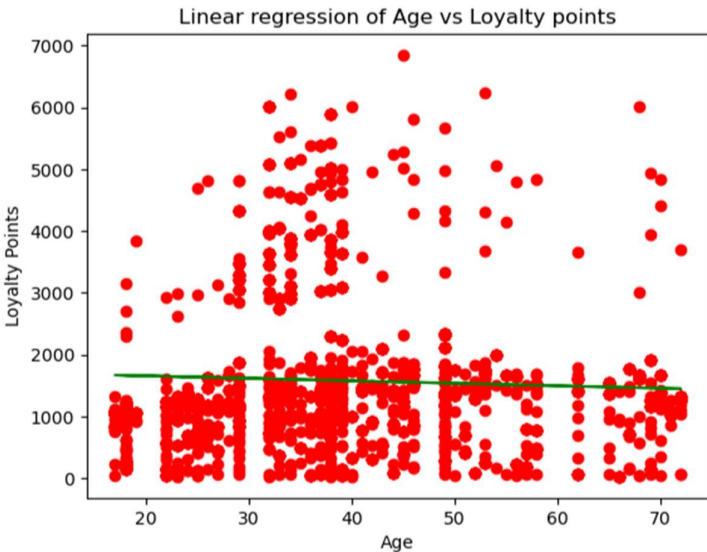


Figure 9 – QQplot for loyalty points

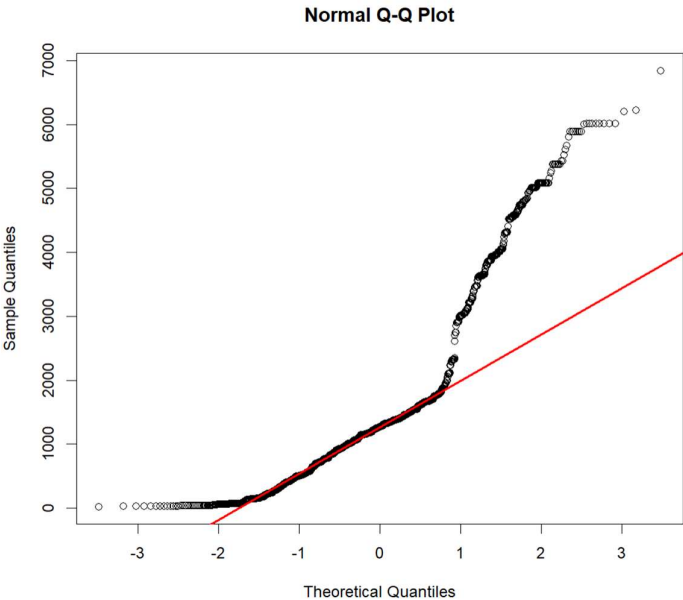


Figure 10 – QQplot for age

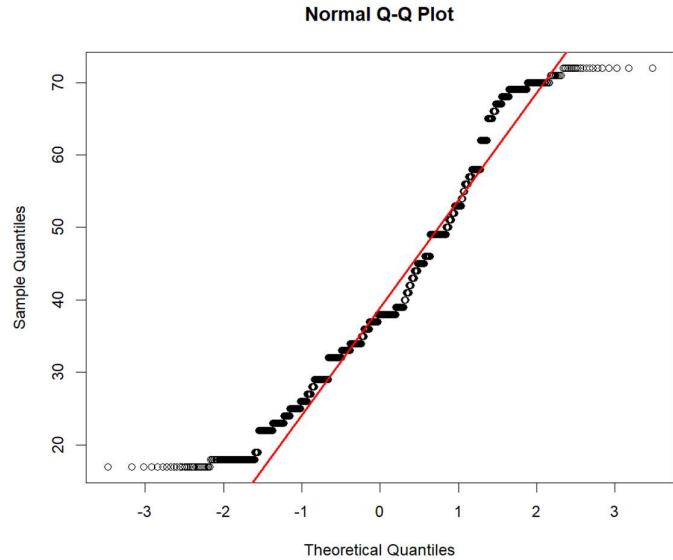


Figure 11 – QQplot for spending score

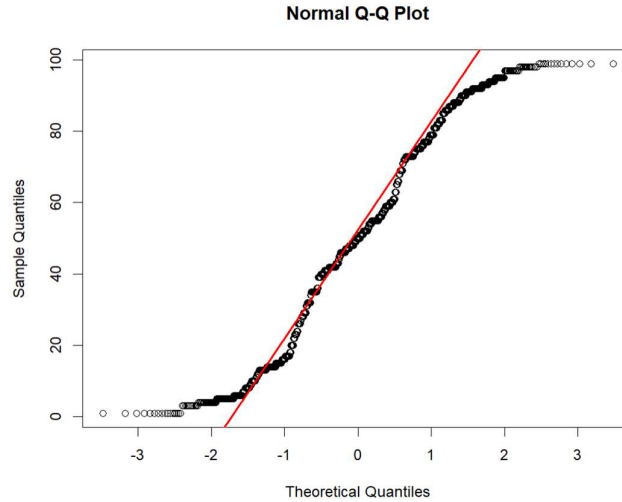


Figure 12 – QQplot for income

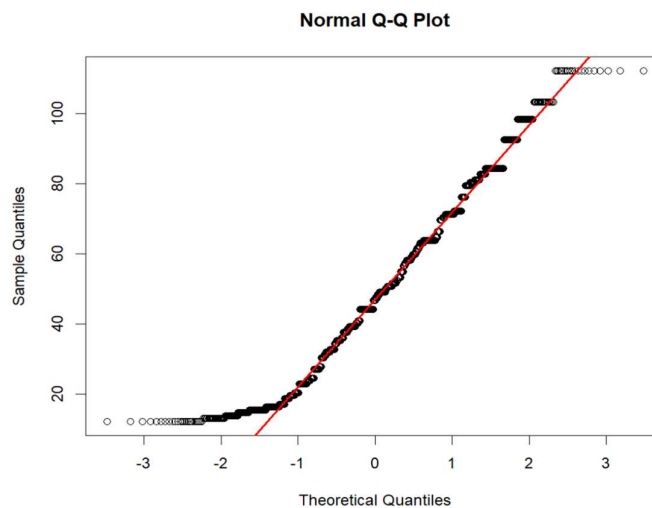


Figure 13 – RStudio console output for Shapiro-Wilk test, skewness and kurtosis for

```
> shapiro.test(turtle1$loyalty_points)

Shapiro-Wilk normality test

data:  turtle1$loyalty_points
W = 0.84307, p-value < 2.2e-16

> skewness(turtle1$loyalty_points)
[1] 1.463694
> kurtosis(turtle1$loyalty_points)
loyalty points [1] 4.70883
```

Figure 14 – RStudio console output for Shapiro-Wilk test, skewness and kurtosis for age

```
> shapiro.test(turtle1$age)

      Shapiro-Wilk normality test

data:  turtle1$age
W = 0.95242, p-value < 2.2e-16

> skewness(turtle1$age)
[1] 0.6088437
> kurtosis(turtle1$age)
[1] 2.80893
```

Figure 15 – RStudio console output for Shapiro-Wilk test, skewness and kurtosis for spending score

```
> shapiro.test(turtle1$spending_score)

      Shapiro-Wilk normality test

data:  turtle1$spending_score
W = 0.96835, p-value < 2.2e-16

> skewness(turtle1$spending_score)
[1] -0.04161713
> kurtosis(turtle1$spending_score)
[1] 2.110333
```

Figure 16 – RStudio console output for Shapiro-Wilk test, skewness and kurtosis for income

```
> shapiro.test(turtle1$income)

      Shapiro-Wilk normality test

data:  turtle1$income
W = 0.96768, p-value < 2.2e-16

> skewness(turtle1$income)
[1] 0.412842
> kurtosis(turtle1$income)
[1] 2.591949
```

Figure 17 – RStudio console output for multiple linear regression summary

```
Call:
lm(formula = loyalty_points ~ age + income + spending_score,
    data = turtle_num)

Residuals:
    Min       1Q   Median       3Q      Max
-1819.11  -350.84    4.61   291.00  1894.62

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2203.0598   52.3609  -42.08  <2e-16 ***
age           11.0607    0.8688   12.73  <2e-16 ***
income       34.0084    0.4970   68.43  <2e-16 ***
spending_score 34.1832    0.4519   75.64  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 513.8 on 1996 degrees of freedom
Multiple R-squared:  0.8399,    Adjusted R-squared:  0.8397
F-statistic: 3491 on 3 and 1996 DF,  p-value: < 2.2e-16
```

Figure 18 – QQplot for residuals of multiple linear regression model

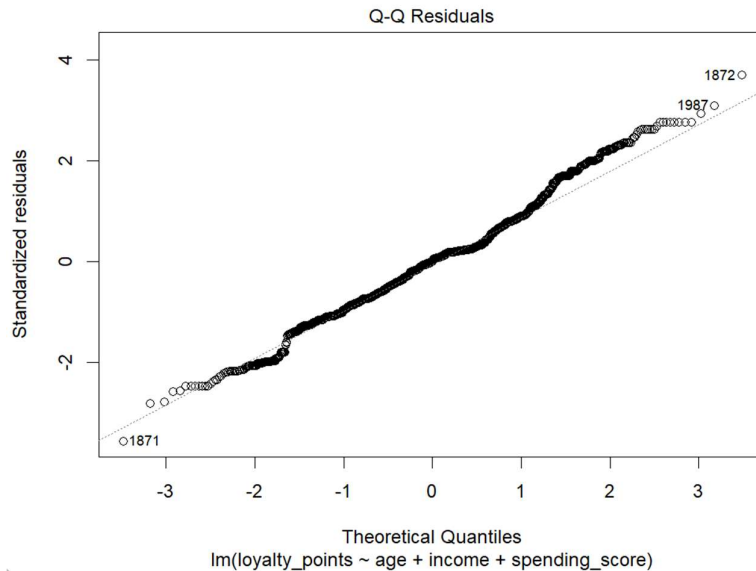


Figure 19 – Residuals vs Fitted plot of multiple linear regression model

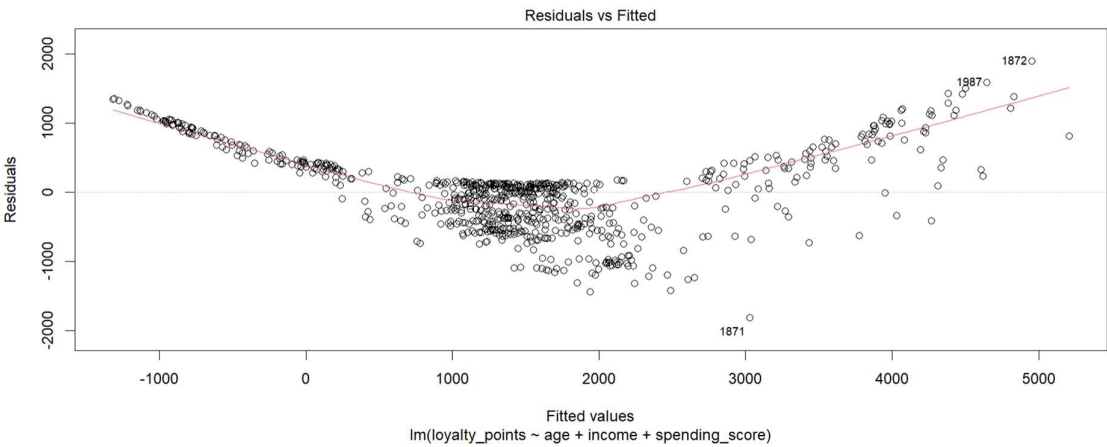


Figure 20 – Data frame showing predictions on synthetic data using multiple linear regression model

	age	spending_score	income	predicted_loyalty_points
1	45	50	40	1364.165
2	50	65	49	2238.292
3	55	70	43	2260.461
4	60	85	70	3746.738

Figure 21 – Unpruned decision on training data

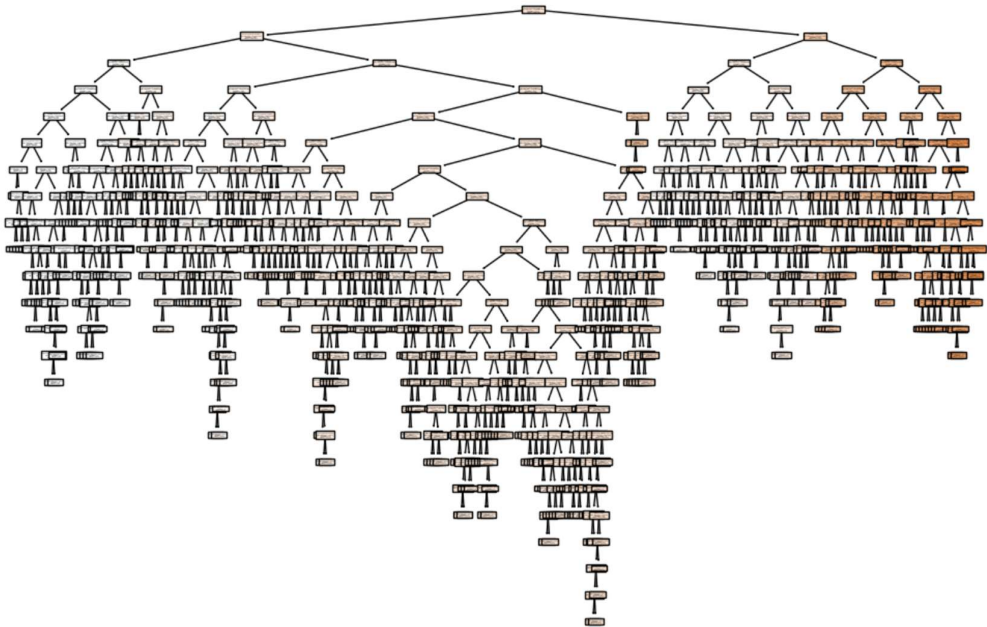


Figure 22 – plot of accuracy values for decision tree

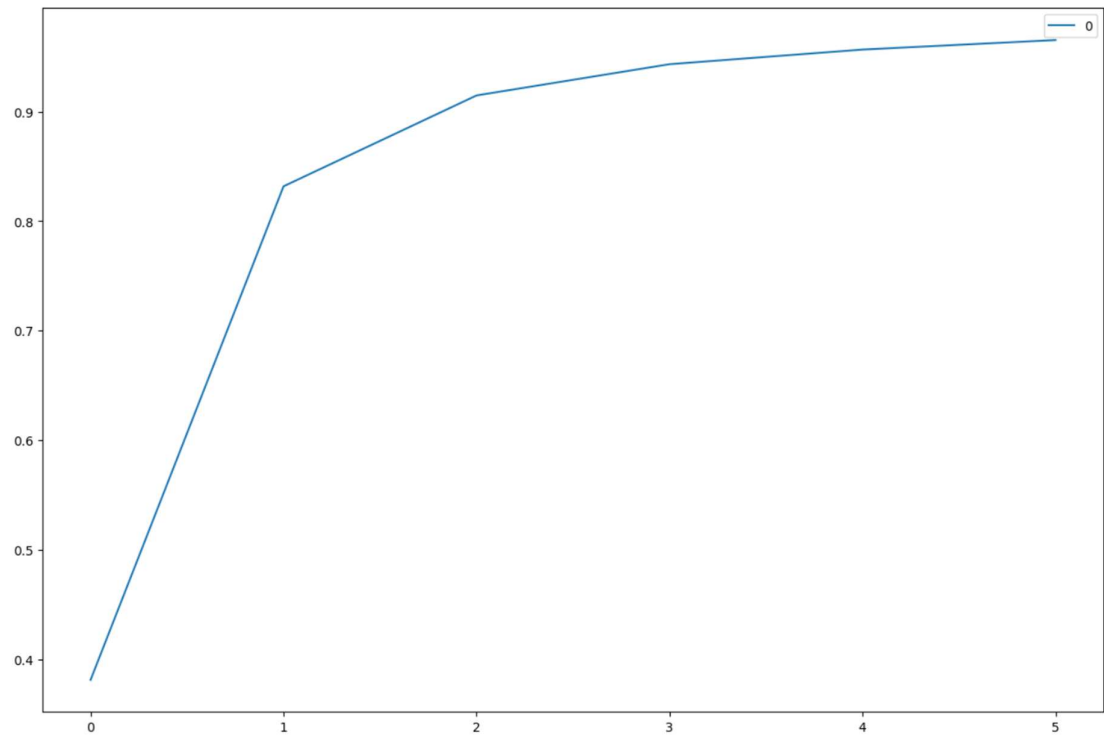


Figure 23 – Pruned decision tree on training data

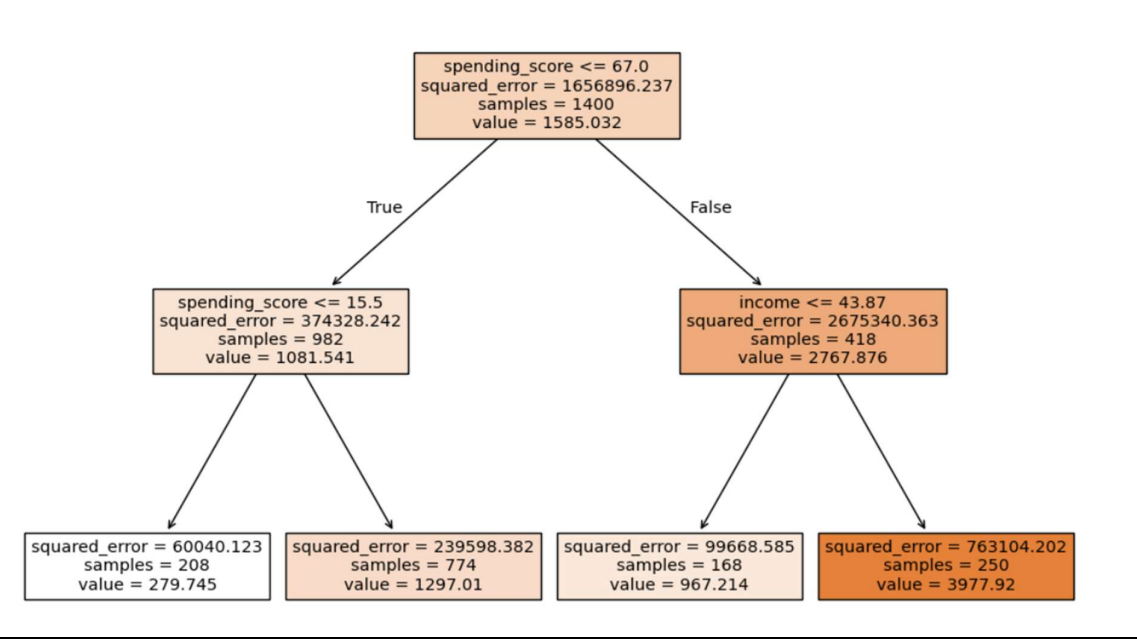


Figure 24 – Python code evaluating errors of pruned tree

```
# Evaluate the pruned tree.  
# Evaluate the model.  
# Predict the response for the data test.  
y_predict_pruned = regressor.predict(X_test)  
  
# Specify to print the MAE and MSE (to evaluate the accuracy of the new model).  
print("Mean Absolute Error: ", metrics.mean_absolute_error(y_test, y_predict_pruned))  
print("Mean Squared Error: ", metrics.mean_squared_error(y_test, y_predict_pruned))  
# Calculate the RMSE.  
print("Root Mean Squared Error: ", math.sqrt(metrics.mean_squared_error(y_test, y_predict_pruned)))
```

Mean Absolute Error: 32.51333333333335
Mean Squared Error: 8072.756666666667
Root Mean Squared Error: 89.84852067044102

```
# Round the pruned tree errors.  
print("Mean Absolute Error: ", round(metrics.mean_absolute_error(y_test, y_predict_pruned), 2))  
print("Mean Squared Error: ", round(metrics.mean_squared_error(y_test, y_predict_pruned), 2))  
# Calculate the RMSE.  
print("Root Mean Squared Error: ", round(math.sqrt(metrics.mean_squared_error(y_test, y_predict_pruned)), 2))
```

Mean Absolute Error: 32.51
Mean Squared Error: 8072.76
Root Mean Squared Error: 89.85

The errors are the same with the original tree and the pruned tree. The model should be used at depth 2 for a simpler model.

Figure 25 – Final decision tree on all of the data

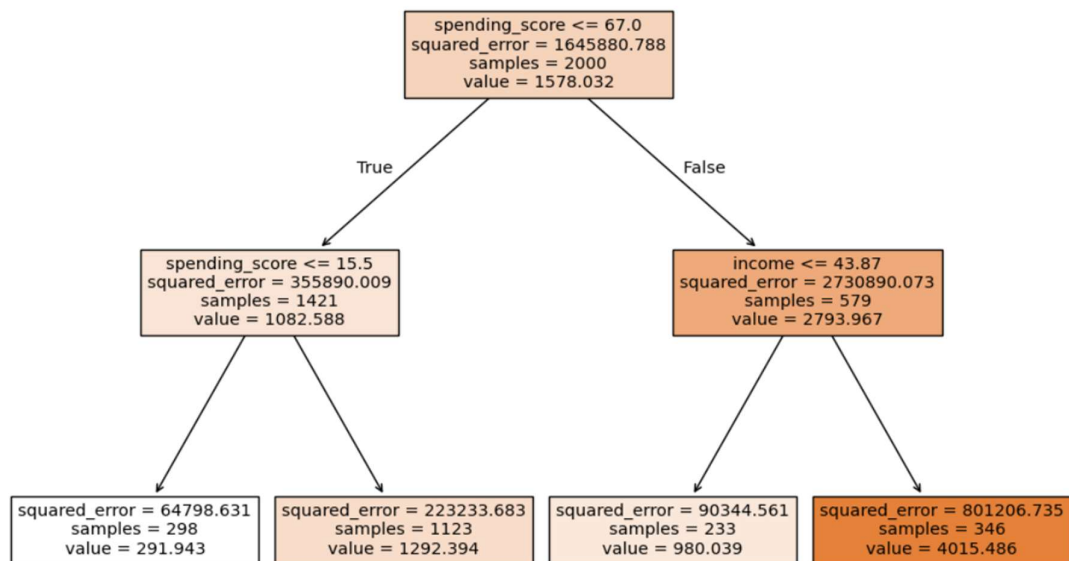


Figure 26 – Pairplot of scatterplots and density plots comparing income and spending score

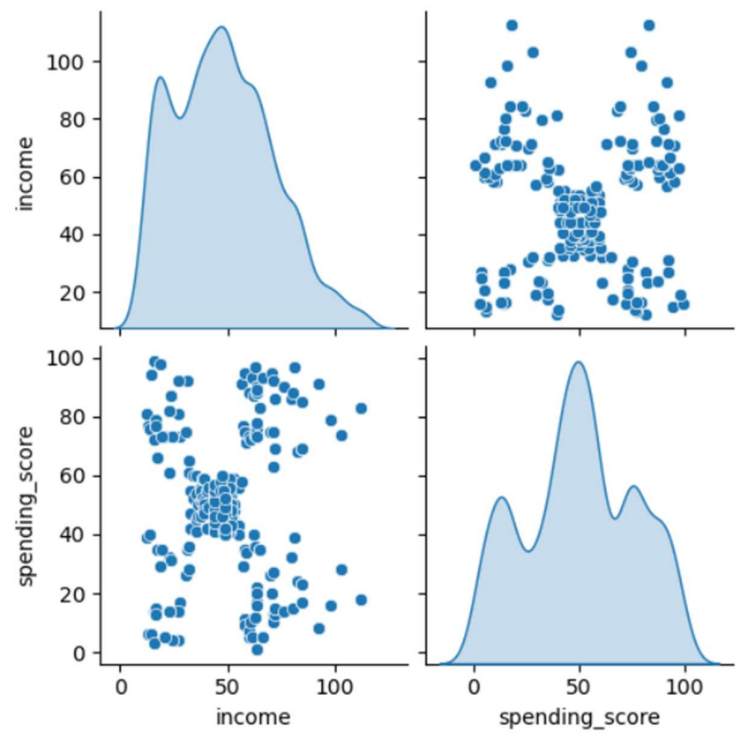


Figure 27 – Silhouette method plot for k-means clustering

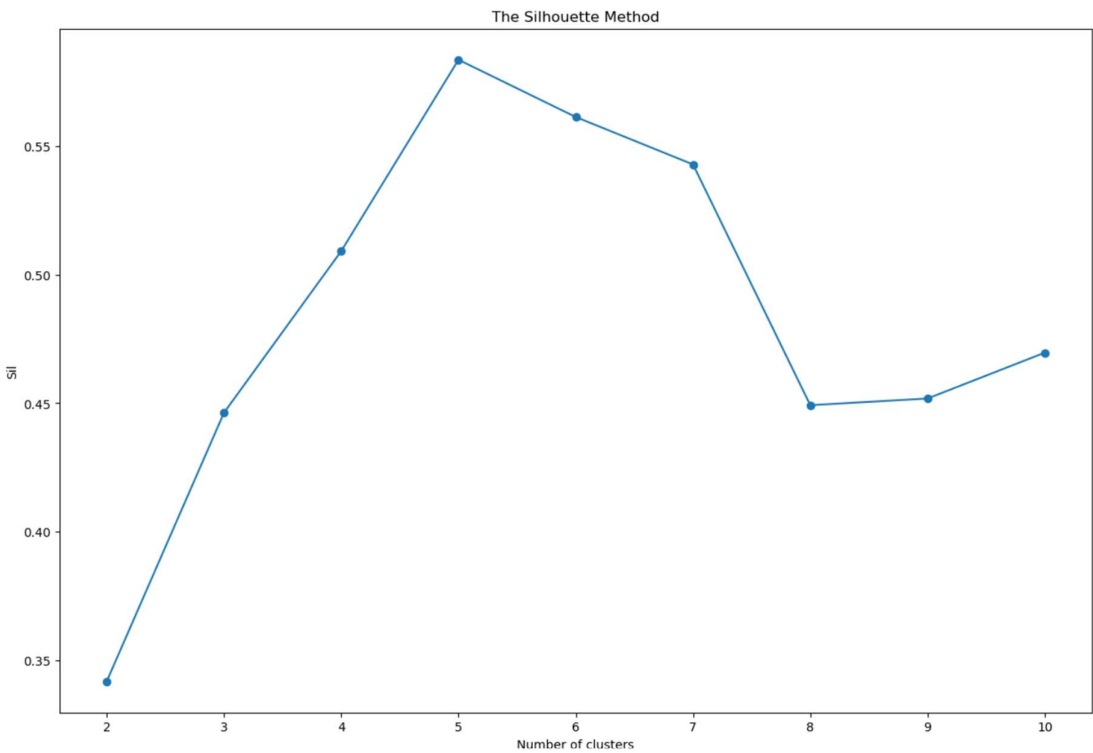


Figure 28 – Elbow method plot for k-means clustering

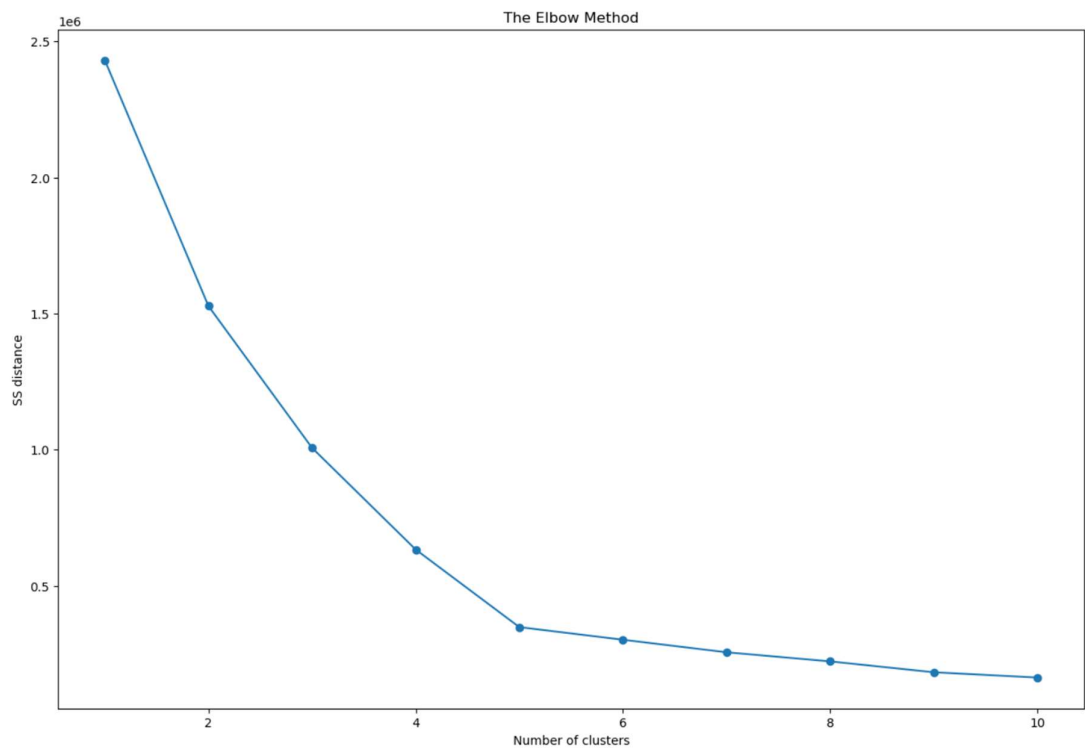


Figure 29 – Scatterplot showing five clusters

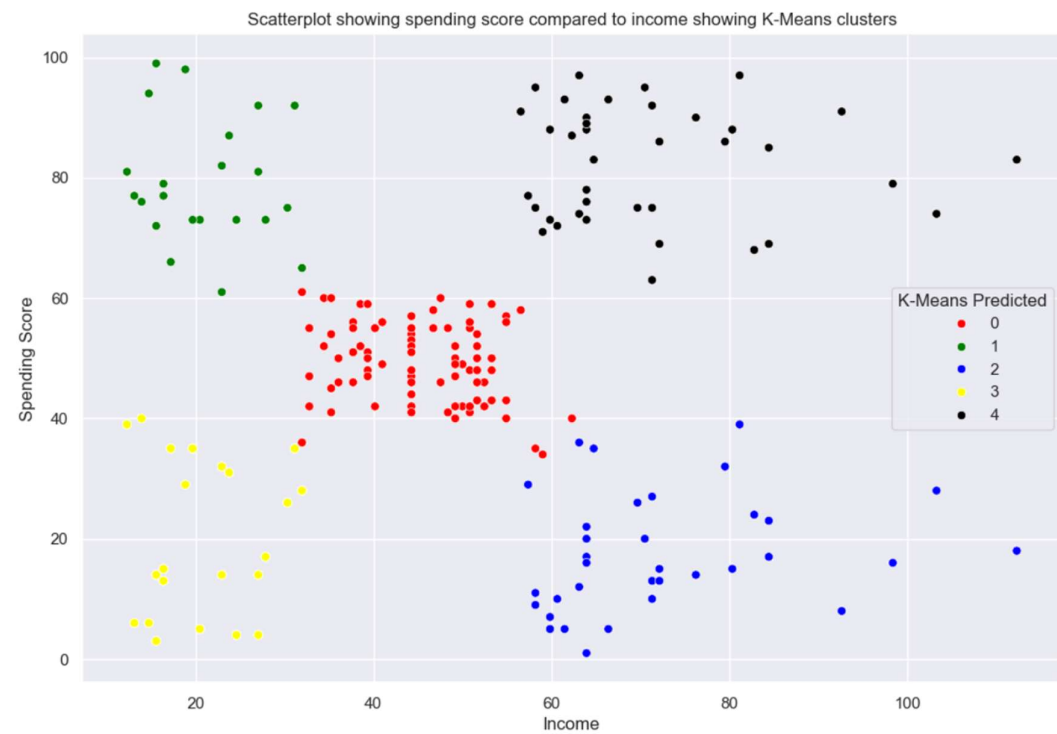


Figure 30 – Bar plot showing 15 most frequent words in reviews

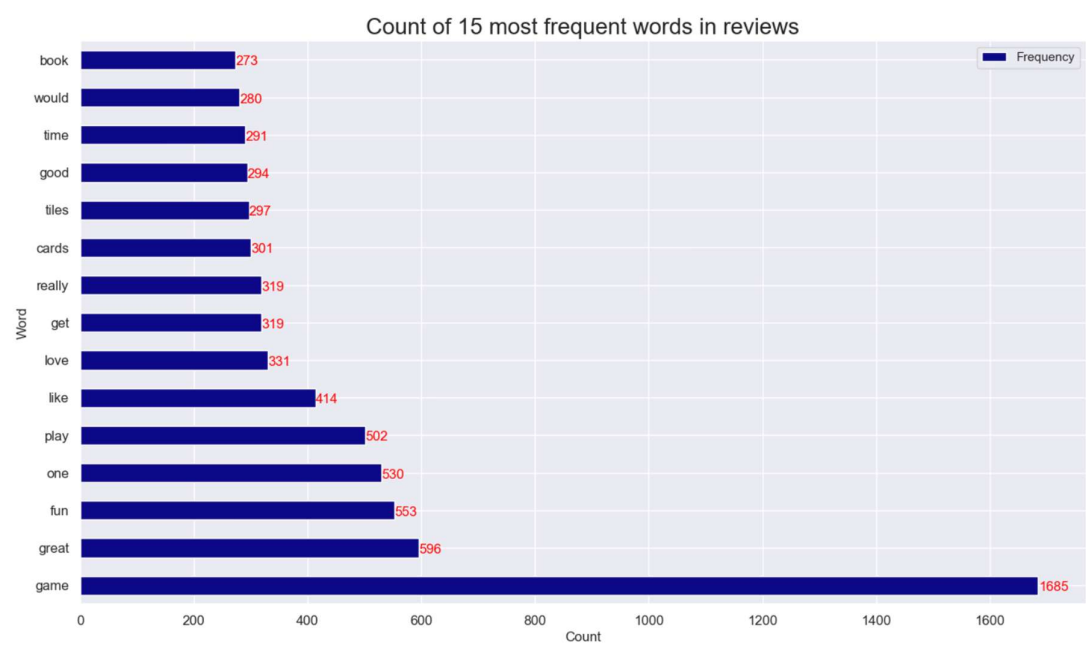


Figure 31 – Bar plot showing 15 most frequent words in summary

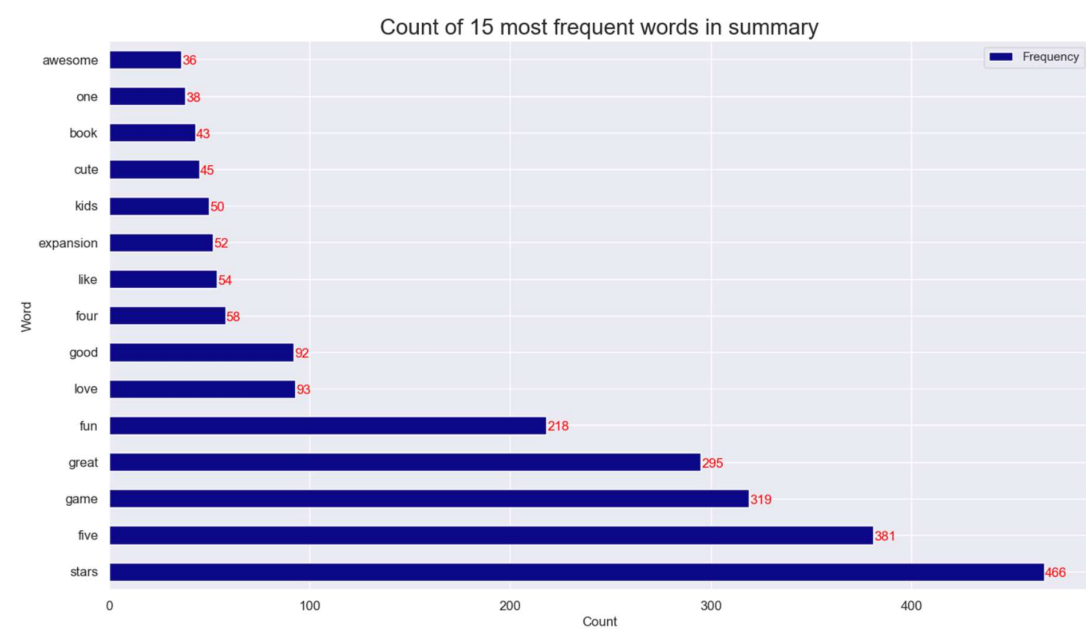


Figure 32 – Histogram showing the distribution of polarity score for reviews

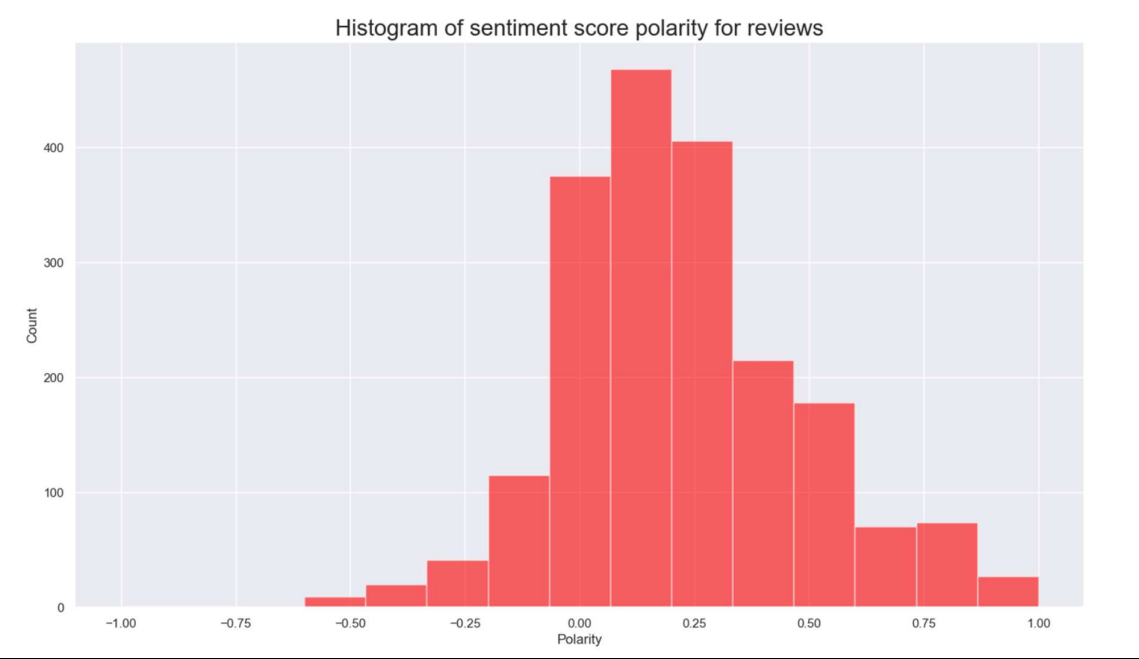


Figure 33 – Histogram showing the distribution of subjectivity scores for reviews

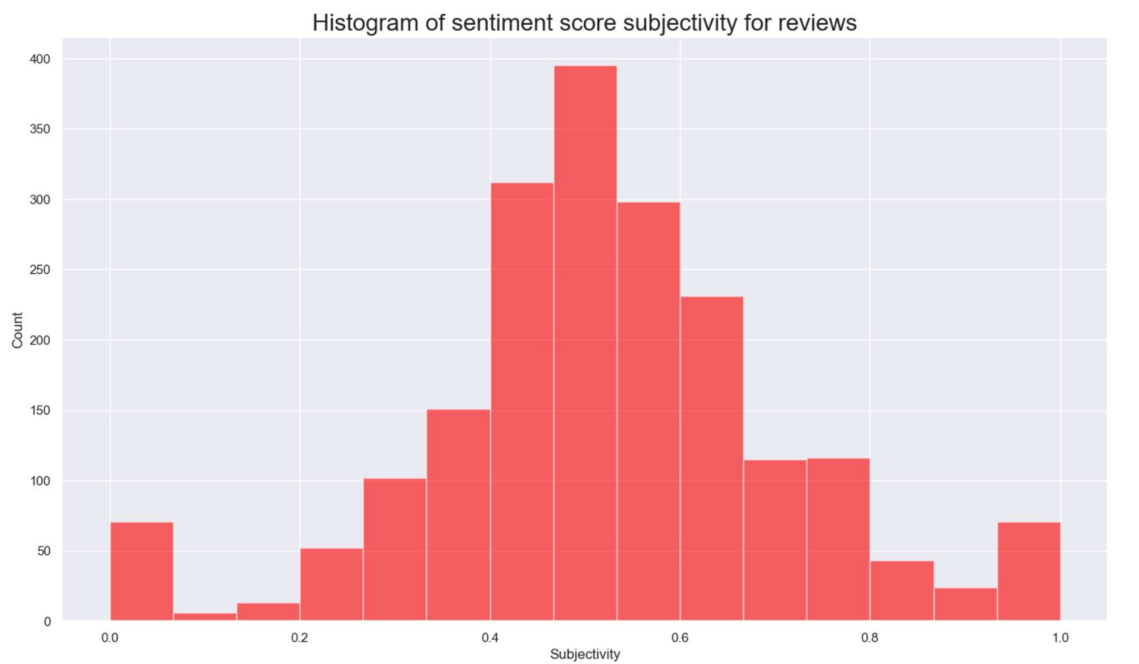


Figure 34 - Histogram showing polarity scores for summary

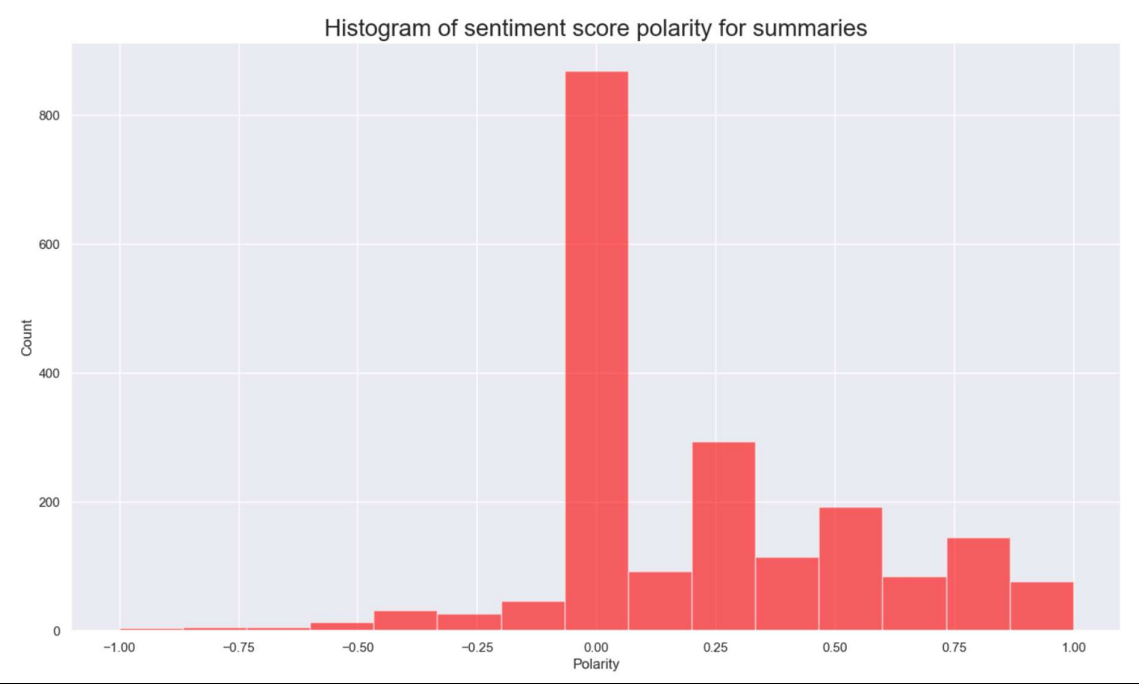


Figure 35 – Histogram showing subjectivity scores for summary

